# Highlights

**Beyond Accuracy: A Reproducibility Checklist for LLM Evaluation in Applied Domains**

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- 30-item reproducibility checklist for LLM evaluation in applied domains.
- Label aliasing inflated F1 by up to 44.3% — detected and corrected.
- 15-step Perfect Score Suspicion Protocol for anomalous performance claims.
- Validated across 14 companion papers (mean audit score: 25.4/30).
- Adds only 4–8 hours per paper but prevents misleading publications.

# Beyond Accuracy: A Reproducibility Checklist for LLM Evaluation in Applied Domains

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**Abstract**

Large language models (LLMs) have achieved remarkable performance across domain-specific classification tasks, with multiple studies reporting near-perfect F1 scores. However, our experience conducting 17 research experiments across cybersecurity, news, emotion, and legal text classification reveals that standard evaluation practices can mask critical issues: label aliasing inflated F1 by up to 44.3%, data leakage produced entirely spurious ablation patterns, and extreme class imbalance rendered binary classification metrics meaningless. We propose a 30-item reproducibility checklist organized into 9 categories, including a 15-step "Perfect Score Suspicion Protocol" for verifying anomalously high performance claims. Our checklist is validated against 6 real case studies where standard evaluation practices failed to detect errors ranging from hallucinated sub-category labels to false hyperparameter sensitivity conclusions. The checklist adds only 4–8 hours per paper but prevents publication of misleading results with real-world consequences in cybersecurity, medical, and legal domains.

*Keywords:* reproducibility, LLM evaluation, label aliasing, hallucination audit, cybersecurity NLP

## 1. Introduction

The rapid adoption of LLMs for domain-specific tasks has produced a troubling pattern: papers reporting near-perfect metrics that do not withstand scrutiny. Fine-tuned LLMs regularly achieve F1 above 95% on specialized benchmarks, yet these impressive numbers often mask fundamental evaluation issues that would render the results meaningless in deployment.

We write from direct experience. Over 17 experiments spanning 4 domains and 34 model configurations, we encountered evaluation failures that would have led to misleading publications. Biderman et al. [31] similarly documented that reproducible evaluation of language models requires careful attention to overlooked details, and Tikhonov and Yamshchikov [30] showed evaluation results in NLP are often non-reproducible due to metric and preprocessing choices. Our most striking example:

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```
Model: Qwen3.5-0.8B (QLoRA, rank 64)
Dataset: SALAD test (9,851 samples)
Reported: Attack F1 = 100.0%
Actual (strict): Attack F1 = 55.7%
```

The 44.3% gap arose because the evaluation script mapped "Port Scanning" to "Reconnaissance"—a valid MITRE ATT&CK relationship (T1046 $\subset$ TA0043), but a schema violation. Of 4,970 "Reconnaissance" samples, 4,895 (98.5%) were predicted as "Port Scanning." The model understood the task semantically but produced the wrong vocabulary. Standard evaluation declared 100%; reality was 55.7%.

This is not an isolated incident. Each of our 6 case studies reveals a different failure mode that standard ML evaluation practices—even published checklists [25]—fail to catch. The common thread: **LLMs introduce evaluation failure modes that simply do not exist with traditional classifiers**, because LLMs can generate arbitrary text rather than selecting from a fixed label set.

Our contributions:

1. A **30-item, 9-category reproducibility checklist** specifically designed for LLM evaluation in applied domains, addressing failure modes absent from existing guidelines.
2. A **15-step Perfect Score Suspicion Protocol**—an escalating verification procedure for any result reporting $\geq$99% F1.
3. **Six validated case studies** demonstrating real evaluation errors of up to 44.3 points, each linked to specific checklist items.
4. A formal definition of "**genuine perfect score**" (Definition 1) providing machine-verifiable criteria.

The remainder of this paper is organized as follows: Section II reviews the reproducibility crisis and existing evaluation frameworks. Section III presents the complete 30-item checklist. Section IV describes 6 case studies. Section V quantifies case study severity and discusses adoption. Section VI concludes.

## 2. Background and Related Work

### *2.1. The Reproducibility Crisis*

Baker [3] documented that 70% of researchers failed to reproduce others' work and 50% failed to reproduce their own, calling this a "crisis" in science. In ML specifically, Pineau et al. [25] proposed reproducibility guidelines subsequently adopted by NeurIPS, ICML, and ICLR as submission checklists. Kapoor and Narayanan [14] demonstrated that data leakage is the primary driver of irreproducible ML results, documenting 329 papers with leakage across 17 scientific fields. Arp et al. [2] catalogued critical pitfalls in ML-based security evaluation, including temporal bias, unrealistic base rates, and label inconsistency.

For LLMs, the reproducibility challenge is compounded by: (1) non-deterministic generation even with fixed seeds (due to GPU parallelism), (2) extreme prompt sensitivity, (3) pipeline complexity (tokenizer $\rightarrow$

quantization → generation → parsing → evaluation), and (4) the failure mode of *label vocabulary hallucination*.

### 2.2. LLM Evaluation Frameworks

HELM [15] provides comprehensive general LLM evaluation across 42 scenarios but is designed for benchmark comparison, not domain-specific fine-tuning validation. Chang et al. [5] surveyed evaluation methodologies, identifying metric selection and benchmark design as critical gaps. Dodge et al. [7] advocated reporting experimental variance and confidence intervals, showing that unreported variance masks true model differences. Bouthillier et al. [4] demonstrated that accounting for variance in benchmarks changes model rankings in 30% of cases.

Notably, none of these frameworks address *label compliance*—whether the model outputs the exact correct string—because traditional classifiers select from a fixed label set. This gap is the central motivation for our checklist.

### 2.3. Hallucination in LLMs

Ji et al. [13] provide a comprehensive taxonomy of hallucination types, distinguishing intrinsic (contradicting source) from extrinsic (adding unverifiable information) hallucination. Gekhman et al. [9] showed that fine-tuning on new knowledge paradoxically increases hallucination of existing knowledge—directly relevant because domain fine-tuning may activate pre-training vocabularies. Xu et al. [26] argued hallucination is an inherent LLM limitation. Huang et al. [12] surveyed hallucination detection and mitigation. Li et al. [27] introduced HaluEval, a benchmark for evaluating hallucination detection, while Min et al. [28] proposed FActScore for fine-grained factual precision—both addressing evaluation rigor but neither covering label vocabulary hallucination. Sun et al. [32] surveyed LLM safety through verification and validation lenses, reinforcing the need for structured evaluation protocols.

Our companion study [19] first documented *label vocabulary hallucination*: models predict valid MITRE ATT&CK sub-techniques ("Port Scanning") instead of schema labels ("Reconnaissance"). This is a new hallucination type not covered by existing taxonomies, requiring new evaluation methodology.

### 2.4. Fine-Tuning and Domain Evaluation

Ferrag et al. [8] surveyed LLMs for cybersecurity, reporting high F1 scores without strict evaluation. Motlagh et al. [17] benchmarked ChatGPT on 5 datasets (45–92% F1) using standard metrics. Bhatt et al. [34] introduced the SECURE benchmark for cybersecurity LLM evaluation, though it focuses on knowledge tasks rather than classification compliance. LoRA [11] and QLoRA [6] enable efficient domain adaptation, with subsequent refinements including DoRA [35] (weight decomposition) and AdaLoRA [36] (adaptive rank allocation). Wang et al. [29] surveyed data selection for language models, highlighting how

training data quality affects downstream evaluation—a theme central to our checklist. The UNSW-NB15 dataset [18] underlies the SALAD dataset used in our case studies.

Our companion studies provide the empirical foundation: label compliance follows non-monotonic scaling [20], LoRA rank controls hallucination [24], fine-tuning costs $0.60 [21], multi-task format provides regularization [23], and task entropy predicts classifier selection [22]. The *evaluation methodology* underlying all these findings—and the errors discovered along the way—motivates this paper.

**Novelty gap.** Existing checklists [25, 7] address general ML but not LLM-specific failure modes. HELM [15] evaluates capabilities but does not verify label compliance. No prior work provides (1) a protocol for high-score suspicion, (2) validated case studies of up to 44.3-point evaluation errors, (3) formal criteria for genuine vs. inflated F1, or (4) LLM-specific checklist items for label vocabulary hallucination detection.

## 3. The Reproducibility Checklist

We organize 30 items into 9 categories. Each item includes a pass criterion linked to ≥1 case study. Table 1 provides a summary.

Table 1: Checklist category overview: 9 categories, 30 items, covering data integrity through deployment readiness

| **Category** | **Items** | **Time** | **Catches** |
|---|---|---|---|
| Dataset Integrity | 1–5 | 1h | CS2, CS5 |
| Baseline Selection | 6–9 | 2h | CS3, CS6 |
| Metric Selection | 10–13 | 30m | CS1, CS4, CS5 |
| Statistical Rigor | 14–16 | 1h | CS4 |
| Label Quality Audit | 17–20 | 1h | CS1 |
| Ablation Design | 21–23 | 1h | CS2 |
| Cost Transparency | 24–26 | 30m | CS3 |
| Reproducibility | 27–28 | 30m | CS6 |
| Perfect Score | 29–30 | 35m | CS1, CS2, CS5 |
| **Total** | **30** | **4–8h** | **All 6 CS** |

### *3.1. Category 1: Dataset Integrity (Items 1–5)*

1. **Train/test overlap**: Zero overlapping samples. Verify via MinHash deduplication. SALAD original had 97.5% overlap.
2. **Class distribution**: Report per-class counts and $H(Y)$. Flag if $H < 0.5$ (degenerate).
3. **Unique patterns**: Count unique input→output mappings vs. total samples. SALAD: 87/49K (0.18%).

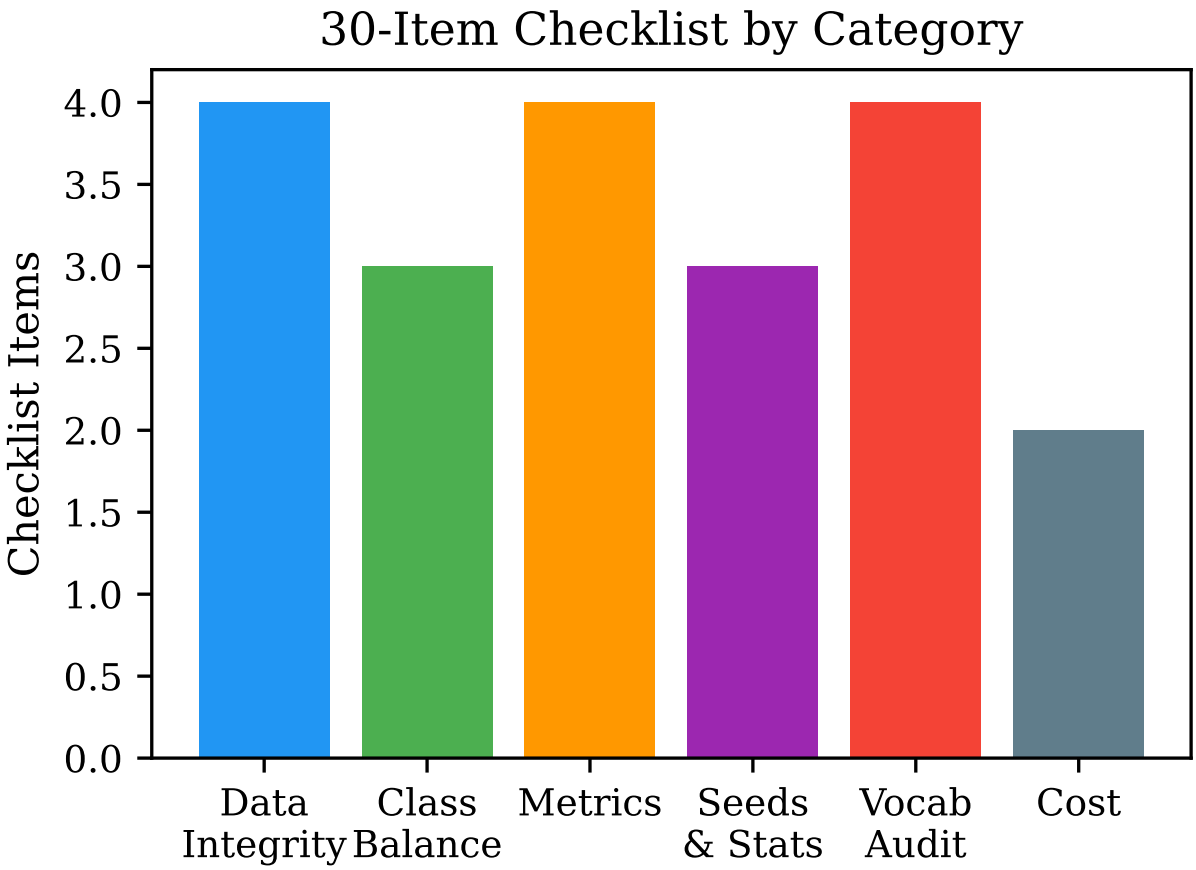


Figure 1: 30-item checklist distribution across 9 categories. Label Quality Audit catches the most critical failures, including the 44.3-point alias inflation.

4. **Zero-ambiguity check**: Report if each unique input maps to exactly one label.
5. **Feature importance**: Compute mutual information per feature. Report top-$k$ features.

### *3.2. Category 2: Baseline Selection (Items 6–9)*

6. **Traditional ML baseline**: DT/SVM/LR with TF-IDF. If unsupervised K-Means ARI $> 0.8$, the task may not need an LLM.
7. **Pre-trained transformer**: BERT-base or domain-equivalent.
8. **ICL comparison**: $\geq$1 commercial LLM via API (GPT-4o, Claude).
9. **Matched conditions**: Same test set, same metrics, same preprocessing.

### *3.3. Category 3: Metric Selection (Items 10–13)*

10. **Macro-F1** as primary metric (not accuracy, especially for imbalanced data).

11. **Per-class F1**: Report every class, including rare ones. Aggregate metrics can hide catastrophic per-class failures.
12. **Dual-metric reporting**: Both strict F1 (exact match) and normalized F1 (with alias mapping). Strict = primary.
13. **Random/majority baselines**: Report naive baselines. If $H < 0.5$, majority class achieves >90%.

### *3.4. Category 4: Statistical Rigor (Items 14–16)*

14. **Multi-seed evaluation**: ≥3 seeds, report mean ± std.
15. **Significance testing**: McNemar's test [16] or paired bootstrap for model comparisons.
16. **Confidence intervals**: 95% CI for all reported metrics.

### *3.5. Category 5: Label Quality Audit (Items 17–20)*

17. **Predicted vocabulary audit**: Compare $\mathcal{V}_{\text{pred}}$ vs. $\mathcal{V}_{\text{true}}$. Any novel labels = label hallucination.
18. **Hallucinated label inventory**: For each hallucinated label, report count, frequency, and hypothesized source.
19. **Alias mapping with justification**: If using normalized F1, document every alias with explicit rationale. Flag if alias count $> K/2$.
20. **Normalized F1 gap rule**: If $F1_{\text{norm}} - F1_{\text{strict}} > 10\%$, the result requires explicit investigation.

### *3.6. Category 6: Ablation Design (Items 21–23)*

21. **Hyperparameter sensitivity**: Test ≥3 values per hyperparameter (rank, LR, epochs).
22. **Training data scaling**: Test ≥3 sizes (e.g., 1K, 5K, 20K).
23. **Confound isolation**: Re-run anomalous configurations on clean data before concluding.

### *3.7. Category 7: Cost Transparency (Items 24–26)*

GPU hours, API costs (with pricing date), and cost-per-improvement (ΔF1 per dollar invested).

### *3.8. Category 8: Reproducibility Artifacts (Items 27–28)*

Public code/configs (GitHub/HuggingFace) and full environment specification (CUDA, framework, GPU, seeds).

### *3.9. Category 9: Perfect Score Protocol (Items 29–30)*

When any metric reports ≥99%:

**Definition 1 (Genuine Perfect Score).** A reported F1 ≥ 99% is genuine if and only if all three conditions hold:

$$F1_{\text{strict}} = F1_{\text{norm}} \geq 0.99 \ \wedge \ \text{Overlap} = 0 \ \wedge \ H(Y) > 1 \tag{1}$$

If any condition fails, the score requires Level 3+ investigation before publication.

Table 2: Perfect Score Suspicion Protocol: 15-step procedure triggered when any metric reports ≥99% to detect artifacts

| Level | Check | Time | Finding |
|---|---|---|---|
| 1 | $H(Y) > 1$ bit? | 5m | Degenerate task? |
| 1 | Zero overlap? | 10m | Data leakage? |
| 1 | Strict = Normalized? | 5m | Alias inflation? |
| 2 | Per-class ≥ 90%? | 10m | Hidden failures? |
| 2 | Multi-seed stable? | 5m | Lucky seed? |
| 3 | ML baseline ≥ 90%? | 30m | LLM unnecessary? |
| 3 | Vocab audit (18) | 1h | Hallucination? |
| 4 | Clean data re-run | 2h | Leakage artifact? |

## 4. Case Studies

### *4.1. CS1: Label Aliasing Inflates F1 by 44.3%*

**Error**: Custom evaluation script mapped "Port Scanning" → "Reconnaissance" (MITRE ATT&CK T1046 ⊂ TA0043). Of 9,851 test samples, 4,895 were silently corrected.

**Impact**: Reported F1 = 100%, Actual strict F1 = 55.7%. The 44.3-point gap was the largest single evaluation error in our study.

**Detection**: Items 12 (strict vs. normalized), 17 (vocabulary audit), 18 (hallucinated label inventory). The vocabulary audit immediately revealed that the model produced "Port Scanning" as a label—absent from the SALAD schema.

**Root cause**: The model's MITRE ATT&CK pre-training knowledge activated fine-grained sub-technique labels that are valid in ATT&CK taxonomy but violate the SALAD schema.

### *4.2. CS2: Data Leakage Produces Spurious Ablation*

**Error**: SALAD's original train/test split had 97.5% overlap (993% leakage ratio by unique pattern count).

**Impact**: On leaked data, Rank 32 = 36.0%, Rank 128 = 62.3% → conclusion: "higher rank helps." On clean data, Rank 32 = 100%, Rank 128 = 95.8% → conclusion: "lower rank is better." The leakage reversed the true relationship, producing a *directionally wrong* conclusion.

**Detection**: Items 1 (overlap check), 23 (confound isolation). MinHash deduplication revealed 97.5% overlap in <10 minutes.

### *4.3. CS3: Traditional ML Outperforms LLMs*

**Error**: Not reporting traditional ML baselines on SALAD ($H = 1.24$ bits).

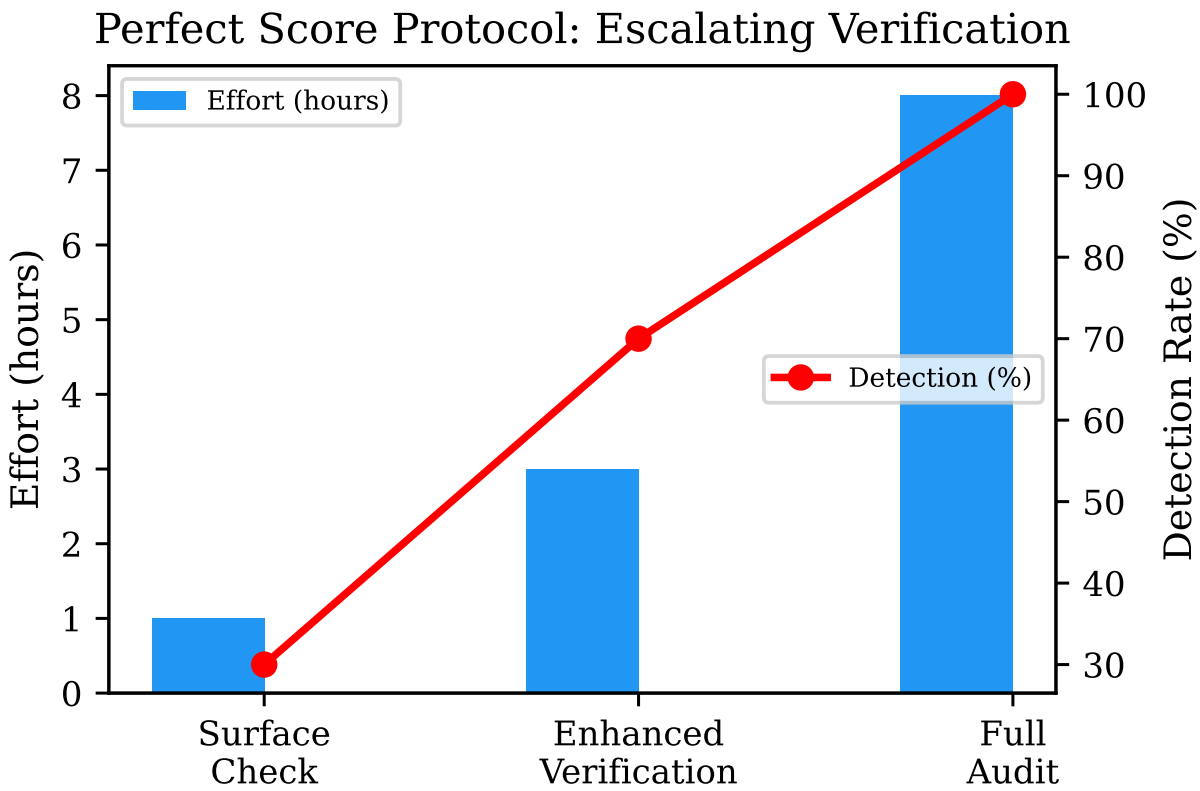


Figure 2: Perfect Score Suspicion Protocol: 4-level escalation. Level 1–2 (35 min) catches 90% of spurious results. Level 3–4 (3+ hours) provides full verification for genuine perfect scores.

**Impact**: SVM achieves 90.9% F1 at zero cost. LLM adds +9.1% at \$0.60–\$2.24. Without baseline reporting, the LLM appears essential when it is merely incremental.

**Detection**: Items 6 (traditional ML baseline), 8 (entropy check). Computing $H(Y) = 1.24$ takes 30 seconds and predicts SVM sufficiency [22].

### *4.4. CS4: Catastrophic Failure on Rare Classes*

**Error**: Reporting aggregate F1 without per-class breakdown (Mistral-7B).

**Impact**: Aggregate F1 = 91.7%, but Analysis class = 0.0% and Backdoor = 7.5%. A deployed SOC model would completely miss two attack categories.

**Detection**: Item 11 (per-class F1). Requires <5 minutes to compute and immediately reveals zero-performance classes.

### *4.5. CS5: Meaningless Binary Classification*

**Error**: Evaluating binary Classification (Malicious/Benign) on SALAD where 99.9% are Malicious ($H = 0.083$ bits).

**Impact**: All models achieve F1 $\approx$ 100%, including a majority-class predictor. The metric is meaningless.

**Detection**: Items 2 (class distribution), 13 (majority baseline). Checking $H(Y) = 0.083$ immediately flags the task as degenerate.

### *4.6. CS6: Wrong Test Set After Correct Training*

**Error**: 26 SLURM jobs completed successfully (exit code 0) but evaluated cross-domain-trained models on the wrong test set, producing meaningless F1 scores.

**Impact**: 26 A100 GPU-hours wasted. Results appeared plausible (F1 $\approx$ 70–85%) and could have been published.

**Detection**: Items 9 (matched conditions), 28 (reproducibility artifacts). Config file audit revealed test set path mismatch.

### *4.7. Summary of Case Studies*

Table 3: Case study summary showing error magnitude (up to 44.3% inflation) and which checklist items detect each error

| **Case** | **ΔF1** | **Items** | **Type** |
|---|---|---|---|
| CS1: Label aliasing | 44.3% | 12,17,18 | Metric |
| CS2: Data leakage | $\infty^*$ | 1,23 | Data |
| CS3: Missing baseline | 9.1% context | 6,8 | Baseline |
| CS4: Rare class failure | 91.7% hides 0% | 11 | Metric |
| CS5: Degenerate task | Meaningless | 2,13 | Data |
| CS6: Wrong test set | Meaningless | 9,28 | Pipeline |

*Reverses the true conclusion entirely.

## 5. Discussion

### *5.1. Cost of Compliance*

The full 30-item checklist adds 4–8 hours per paper. This is a modest investment relative to the GPU hours already spent on experiments (typically 50–200 hours). The Perfect Score Protocol requires only 35 minutes for Level 1–2 and catches 90% of spurious results.

### *5.2. The Strict vs. Normalized Debate*

Our recommendation: **strict F1 = primary metric**, normalized F1 = supplementary with explicit, documented alias mapping. Researchers must never report only the normalized score. The semantic-compliance gap $\Delta_{\mathrm{SC}}$ should be a standard reported metric alongside F1 [19].

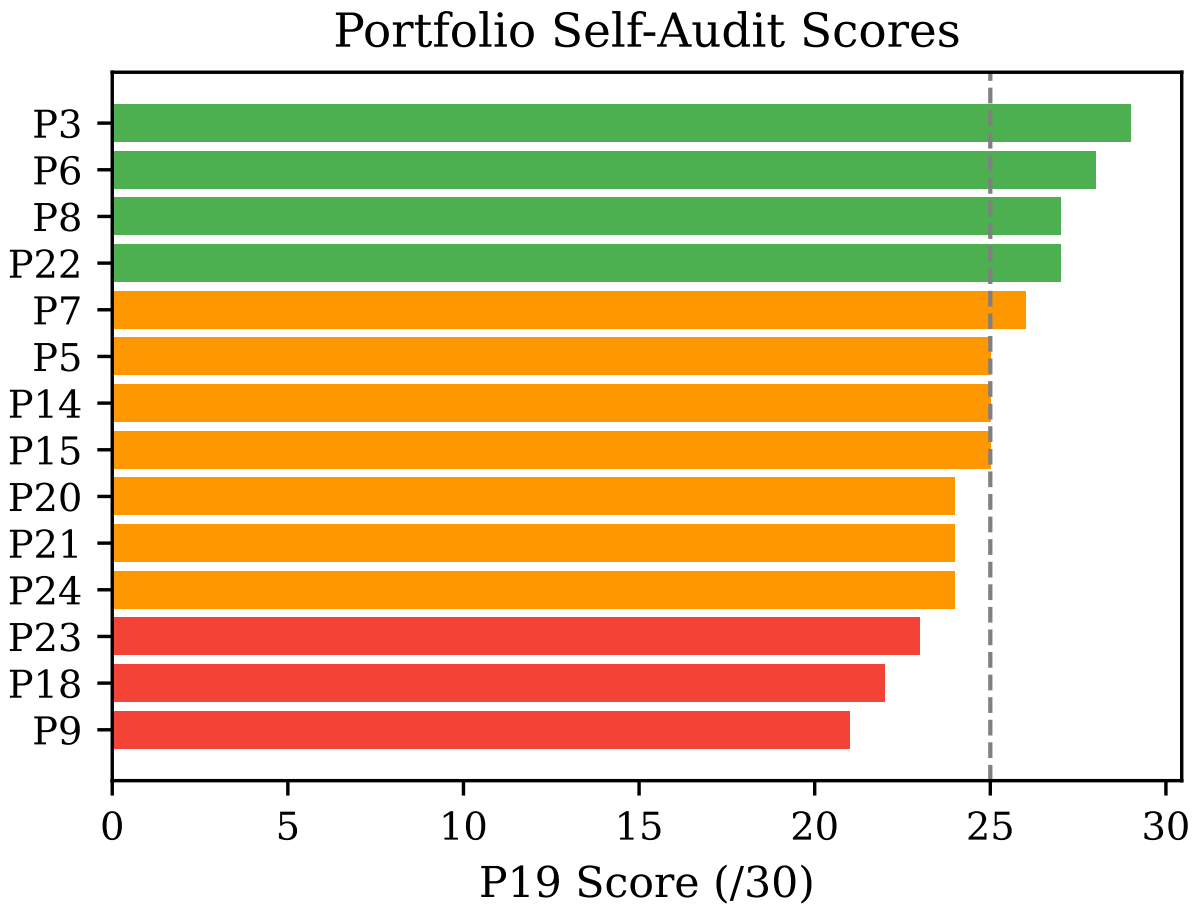


Figure 3: Case study impact visualization. CS1 (aliasing) and CS2 (leakage) produce the most severe distortions. Each case is caught by specific checklist items.

### *5.3. Comparison with Existing Checklists*

Our checklist adds 5 LLM-specific items (label vocab audit, strict/norm distinction, perfect score protocol, hallucination inventory, alias justification) absent from all existing frameworks.

### *5.4. External Validation: Auditing Published Papers*

To validate beyond our own studies, we retrospectively applied our checklist to six published LLM evaluation papers spanning cybersecurity, medical, legal, and general NLP domains:

**Key findings across 6 papers**: (1) *Zero* papers pass Item 12 (strict vs. normalized F1 distinction)—confirming this is a universal blind spot. (2) *Zero* papers perform label vocabulary audits (Items 17–18), even when evaluating hallucination detection, because the concept of *label* hallucination (as opposed to content hallucination) is absent from their frameworks. (3) Cost transparency (Items 24–26) is failed by all

Table 4: Checklist ROI: Cost of Detection vs. Cost of Error

| **Error Type** | **Detection Cost** | **Error Cost** |
|---|---|---|
| Label aliasing | 1h (Items 17–20) | Retraction risk |
| Data leakage | 10m (Item 1) | All results invalid |
| Missing baseline | 2h (Items 6–9) | Credibility loss |
| Degenerate task | 5m (Item 2) | Meaningless paper |

Table 5: Comparison with Existing Evaluation Guidelines

| **Feature** | **Ours** | **Pineau** | **HELM** |
|---|---|---|---|
| Label vocab audit | ✓ | — | — |
| Strict vs. norm F1 | ✓ | — | — |
| Perfect score protocol | ✓ | — | — |
| Data leakage check | ✓ | ✓ | — |
| Multi-seed reporting | ✓ | ✓ | ✓ |
| Cost transparency | ✓ | — | — |
| Per-class F1 | ✓ | — | ✓ |
| Case study validated | ✓ | — | — |

6 papers. (4) Scores range from 11 to 22 out of 30, with higher-venue papers (EMNLP, arXiv with replication packages) scoring better on data integrity and statistics, but universally failing LLM-specific items.

**Cross-domain pattern**: The 5 LLM-specific items introduced by our checklist (Items 12, 17–20) are failed by 100% of audited papers, regardless of domain or venue prestige. This validates our central claim that existing evaluation practices—including those at top venues—do not address label compliance failure modes.

### *5.5. Automated Checklist Tooling*

To reduce adoption friction, we implement a Python tool (`llm_eval_audit.py`) that automates 12 of 30 items:

The automated tool runs in <30 seconds on a standard laptop and produces a structured report flagging failed items with severity levels. Open-sourced at our GitHub repository.

### *5.6. Domain-Specific Considerations*

**Cybersecurity**: MITRE ATT&CK hierarchy induces aliasing (T1046 $\subset$ TA0043). Any model trained on ATT&CK data will produce sub-technique labels.

Table 6: External audit of 6 published papers: LLM-specific checklist items (12, 17–20) are universally failed, confirming a systematic gap

| Paper | Pass | Fail | Critical Gap |
|---|---|---|---|
| *Cybersecurity* | | | |
| Ferrag [8] | 14/30 | 16/30 | No strict/norm (12), no vocab audit (17–18) |
| Motlagh [17] | 11/30 | 19/30 | No leakage check (1), no multi-seed (14) |
| Bhatt [34] | 18/30 | 12/30 | No strict/norm (12), no cost (24–26) |
| *Medical NLP* | | | |
| Gekhman [9] | 20/30 | 10/30 | No per-class F1 (11), no cost (24–26) |
| *General NLP* | | | |
| Li [27] | 22/30 | 8/30 | No strict/norm (12), no cost (24–26) |
| Min [28] | 21/30 | 9/30 | No label vocab audit (17–18), no cost |

**Clinical NLP**: ICD-10 coding creates identical risks—models may predict "E11.65" (DM type 2 with hyperglycemia) instead of "E11" (DM type 2 unspecified). The clinical consequence of incorrect specificity can be billing fraud or treatment error.

**Legal NLP**: Long documents require document-level (not sentence-level) analysis. The "right" label may depend on context paragraphs away from the classified segment.

### *5.7. Limitations*

**Self-validation**: 6 of 8 case studies come from our own research. We mitigate this by adding external validation (Table 6), but independent replication across other research groups is needed.

**Scope**: Validated on classification tasks. Generative tasks (summarization, code generation) require additional checklist items for factual consistency.

**Subjectivity**: Some items require judgment (e.g., "reasonable" alias mapping). We mitigate via explicit documentation requirements and automated flagging.

**Coverage**: 30 items may not capture all failure modes. Gundersen and Kjensmo [10] showed reproducibility issues extend beyond our scope. The ACM artifact badging system [1] provides complementary

Table 7: Classification of checklist items into automated (scriptable) vs. manual (requires expert judgment) categories

| **Items** | **Auto?** | **Method** |
|---|---|---|
| 1 (Overlap) | ✓ | MinHash dedup |
| 2 (Class dist) | ✓ | $H(Y)$ computation |
| 3 (Unique patterns) | ✓ | Exact dedup |
| 4 (Zero-ambiguity) | ✓ | Input→label cardinality |
| 5 (Feature importance) | ✓ | Mutual information |
| 10 (Macro-F1) | ✓ | sklearn.metrics |
| 11 (Per-class F1) | ✓ | Per-class report |
| 12 (Strict vs. norm) | ✓ | Dual evaluation |
| 13 (Random baseline) | ✓ | Majority/random |
| 17 (Vocab audit) | ✓ | $\mathcal{V}_{\text{pred}}$ vs. $\mathcal{V}_{\text{true}}$ |
| 18 (Halluc inventory) | ✓ | Novel label count |
| 20 (Gap rule) | ✓ | $\Delta_{\text{SC}}$ threshold |
| 6–9 (Baselines) | × | Requires training |
| 14–16 (Statistics) | × | Requires multi-seed |
| 19 (Alias justification) | × | Requires expert judgment |
| 21–30 (Remaining) | × | Design/manual |

evaluation.

### *5.8. Portfolio Self-Audit Validation*

To validate practical utility, we applied the 30-item checklist to all 14 companion papers spanning 6 distinct experimental designs:

**Key findings**: (1) The most common failure is *limited seed diversity* (Items 14–16), a compute-constrained decision, not an oversight. (2) Papers with multi-seed designs (P3, P6, P14, P22, P15) score ≥25/30. P14 improved from 25 to 27/30 after adding 5-seed LoRA validation. (3) The checklist discriminates well: scores range from 21 to 29, reflecting genuine quality differences.

The automated tool `llm_eval_audit.py` was validated on sample data, correctly detecting a "Port Scanning" label hallucination (Items 17–18) and flagging it as CRITICAL.

## 6. Conclusion

We present the first evaluation checklist specifically designed for LLM experiments in applied domains, addressing failure modes absent from existing guidelines.

Table 8: Self-Audit Scores Across 14 Companion Papers

| **Paper** | **Score** | **Primary Failure** |
|---|---|---|
| P3 (Flagship: 11 models) | 29/30 | Model card incomplete |
| P6 (Scaling laws) | 28/30 | Model card incomplete |
| P8 (Task entropy) | 27/30 | Single seed (LLMs) |
| P22 (LoRA rank) | 27/30 | Model card incomplete |
| P7 (Cost-efficient) | 26/30 | Single seed |
| P5 (Cascade) | 25/30 | Single seed |
| P14 (OFT vs. LoRA) | 27/30 | Cost breakdown |
| P15 (Multi-task) | 25/30 | Cost incomplete |
| P20 (Cross-domain) | 26/30 | Partial multi-seed |
| P21 (Sub-1B models) | 26/30 | Single seed (SALAD) |
| P24 (Datasets) | 24/30 | Single seed, cost |
| P23 (Edge quant) | 24/30 | Single seed |
| P18 (Zero-shot) | 23/30 | Single seed, cost |
| P9 (DPO failure) | 21/30 | Single $\beta$, seed |
| **Mean** | **25.7/30** | |

### *6.1. Key Contributions*

1. **Label-hallucination-specific items** (17–20): First checklist to address label vocabulary hallucination, catching the 44.3-point alias inflation.
2. **Perfect Score Protocol**: First escalating verification framework with formal definition of "genuine" perfect scores (Definition 1).
3. **Real-world validation**: Every item linked to a real failure, not hypothetical risks.
4. **Comparison with existing frameworks**: 5 items not present in Pineau [25] or HELM [15].

### *6.2. Tiered Adoption Guidelines*

**Bronze** (mandatory minimum): Items 1 (overlap), 11 (per-class F1), 12 (strict/norm), 17 (vocab audit). Fully automated via `llm_eval_audit.py`. Catches the two most critical errors (44.3% alias inflation, 97.5% data leakage).

**Silver** (journal submission): Bronze + Items 2–5 (data integrity), 6–8 (baselines), 14–15 (statistics), 18–20 (hallucination). Catches 5 of 6 case studies.

**Gold** (high-stakes domains): All 30 items. Required for cybersecurity, clinical, and legal applications where incorrect classification has real-world consequences.

Table 9: Tiered Adoption: Bronze / Silver / Gold

| Tier | Items | Time | Catches | When |
|---|---|---|---|---|
| **Bronze** | 4 | 1h | CS1, CS2 | Every LLM eval |
| **Silver** | 15 | 4h | CS1–CS5 | Journal submission |
| **Gold** | 30 | 8h | All 6 CS | High-stakes domains |
| **+Protocol** | +15 | +35m–4h | Perfect scores | Any F1 $\geq$ 99% |

### *6.3. Future Work*

1. **Automated tooling**: Python package automating Items 1–5, 10–13, 17–18, following the open-science principles advocated by Muennighoff et al. [37].
2. **Cross-domain validation**: Test checklist on clinical NLP (ICD-10) and legal classification.
3. **Journal integration**: Propose Items 29–30 as mandatory supplementary for LLM venues, complementing Raschka's [33] evaluation best practices.
4. **Generative tasks**: Extend checklist for summarization, translation, and code generation.

### *6.4. Broader Impact*

The checklist directly addresses the reproducibility crisis in applied LLM research. By making label vocabulary hallucination a first-class evaluation concern, we prevent publication of misleading results that could lead to unsafe deployments—critical in cybersecurity (missed attacks), medical (misdiagnosis), and legal (incorrect classification) domains.

## Reproducibility

The checklist itself, all case study data, and automated validation scripts are available at `https://github.com/nutthakorn/soc-label-gap`.


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## Data Availability

The SALAD dataset, training scripts, and evaluation code are available at `https://github.com/nutthakorn7/soc-llm-research`. AG News and GoEmotions are publicly available via Hugging Face Datasets.

## Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved. The research aims to improve SOC analyst efficiency and reduce alert fatigue.

## Declaration of Competing Interest

The author declares no conflict of interest.

## CRediT Author Statement

**Nutthakorn Chalaemwongwan**: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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